

Lecture 17

ABSOLUTE MOTION ANALYSIS

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ABSOLUTE MOTION ANALYSIS

Today's Objective:

Students will be able to:

1. Determine the velocity and acceleration of a rigid body undergoing **general plane motion** using an absolute motion analysis.



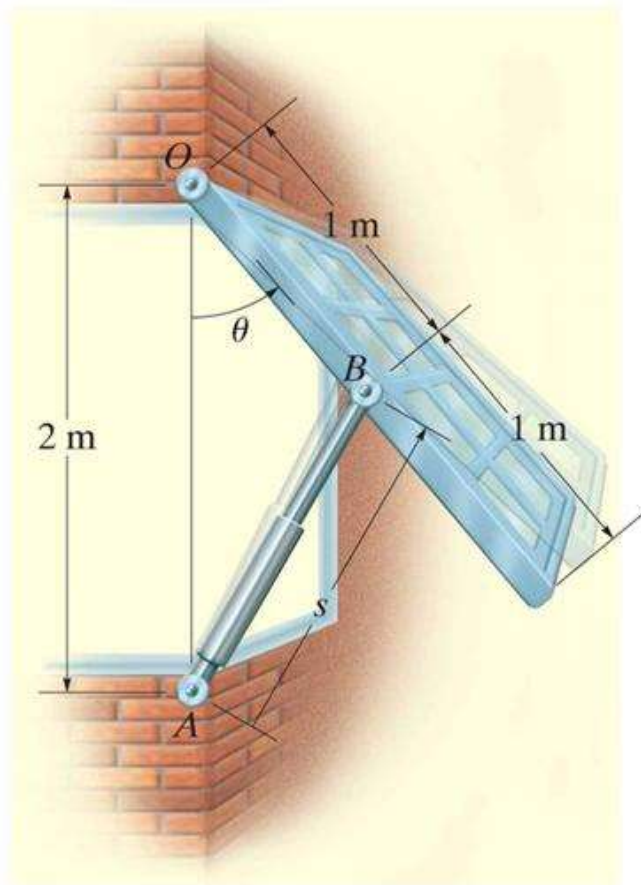
In-Class Activities:

- Reading Quiz
- Applications
- **General Plane Motion**
- Concept Quiz
- Group Problem Solving
- Attention Quiz

READING QUIZ

1. A body subjected to general plane motion undergoes a/an
 - A) translation.
 - B) rotation.
 - C) simultaneous translation and rotation.
 - D) out-of-plane movement.
2. In general plane motion, if the rigid body is represented by a slab, the slab rotates
 - A) about an axis perpendicular to the plane.
 - B) about an axis parallel to the plane.
 - C) about an axis lying in the plane.
 - D) None of the above.

APPLICATIONS

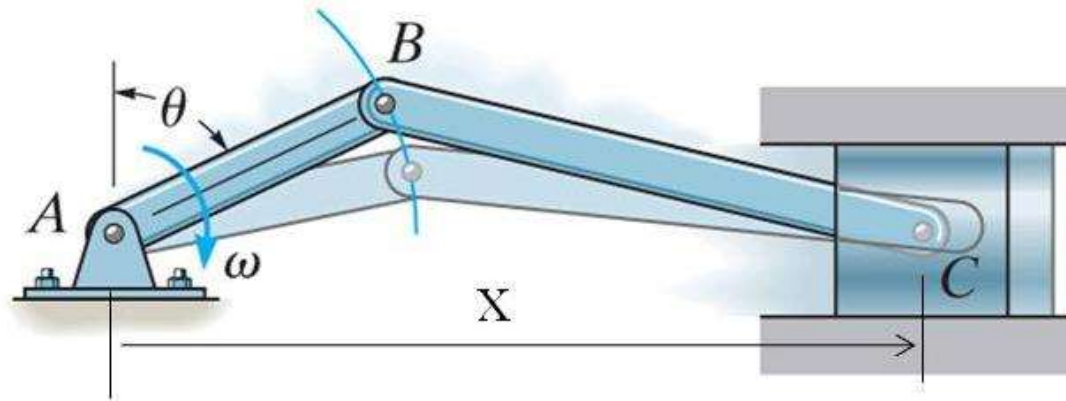


The large window is opened using a hydraulic cylinder AB.

The position B of the hydraulic cylinder rod is related to the angular position, θ , of the window.

A designer has to relate the translational velocity at B of the hydraulic cylinder and the angular velocity and acceleration of the window? How would you go about the task?

APPLICATIONS (continued)

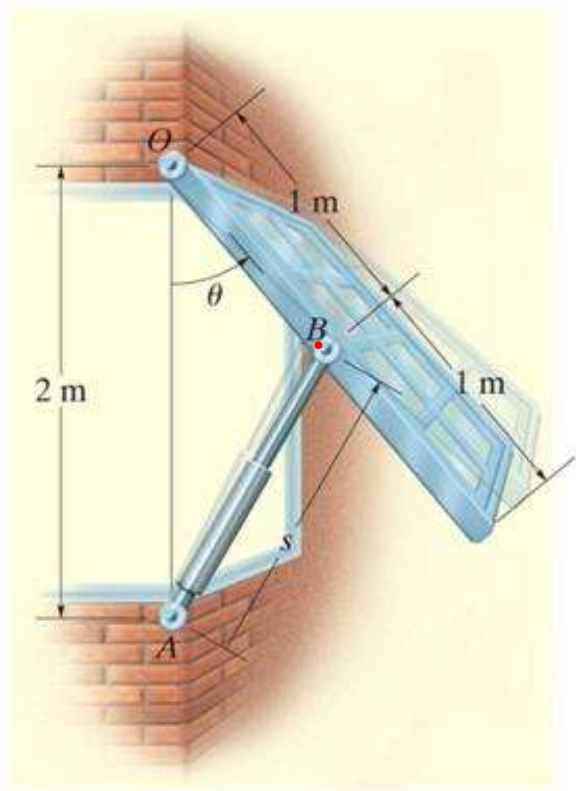


The position of the piston, x , can be defined as a function of the angular position of the crank, θ . By differentiating x with respect to time, the velocity of the piston can be related to the angular velocity, ω , of the crank. This is necessary when designing an engine.

The stroke of the piston is defined as the total distance moved by the piston as the crank angle varies from 0 to 180° . How does the length of crank AB affect the stroke?

ABSOLUTE MOTION ANALYSIS (Section 16.4)

The figure below shows the window using a hydraulic cylinder AB.



The **absolute motion analysis method** relates the position of a point, B, on a rigid body undergoing rectilinear motion to the angular position, θ , of a line contained in the body.

Once a relationship in the form of $s_B = f(\theta)$ is established, the velocity and acceleration of point B are obtained in terms of the angular velocity and angular acceleration of the rigid body by taking the **first and the second time derivatives** of the position function.

Usually, the **chain rule** must be used when taking the derivatives of the position coordinate equation.

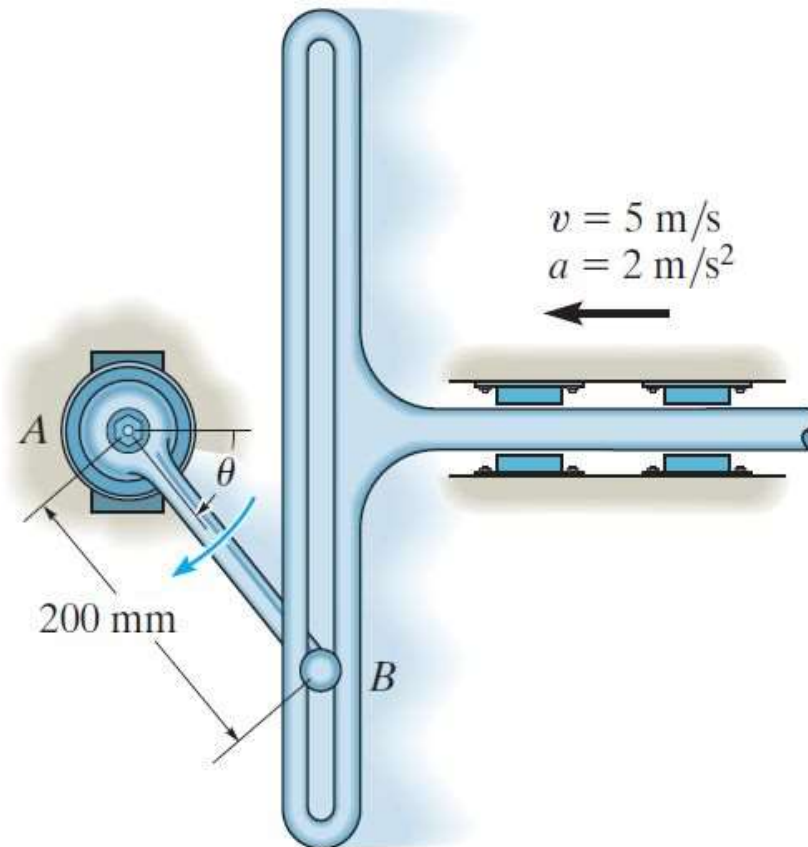
PROCEDURE FOR ABSOLUTE MOTION ANALYSIS

The velocity and acceleration of a point undergoing rectilinear motion can be related to the angular velocity and angular acceleration of a line contained within a body using the following procedure.

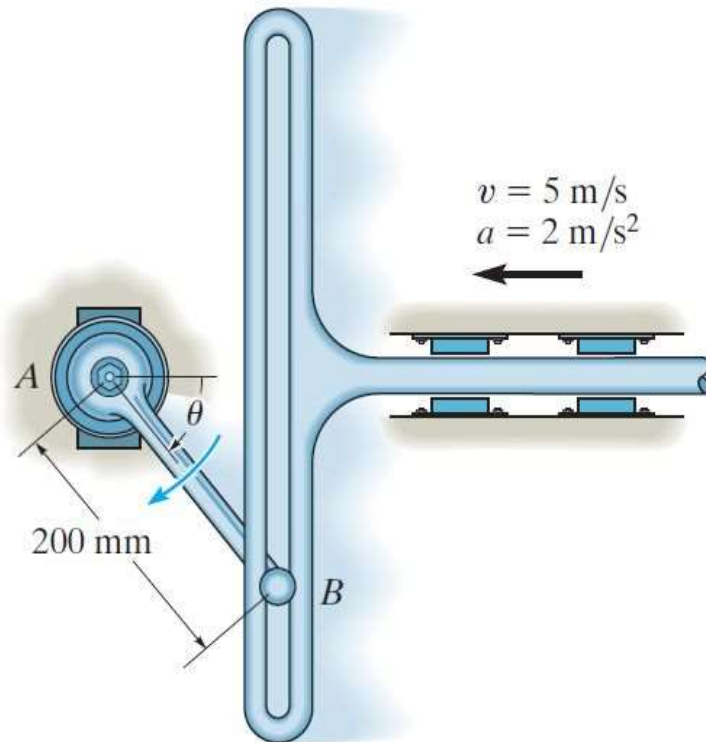
1. Locate point on the body using position coordinate s , which is measured from a fixed origin.
2. From a fixed reference line, measure the angular position θ of a line lying in the body. Using the dimensions of the body, relate s to θ , e.g., $s = f(\theta)$.
3. Take the first-time derivative of $s = f(\theta)$ to get a relationship between v and ω .
4. Take the second time derivative to get a relationship between a and α .

EXAMPLE

At the instant $\theta = 60^\circ$, the slotted guide rod is moving to the left with an acceleration of 2 m/s^2 and a velocity of 5 m/s . Determine the angular acceleration and angular velocity of link AB at this instant. **(Problem No. 16-40)**



EXAMPLE



Given: Link AB is rotating clockwise direction. At the instant $\theta = 60^\circ$, the slotted guide rod is moving to the left with $a = 2 \text{ m/s}^2$ and $v = 5 \text{ m/s}$.

Find: The angular velocity and angular acceleration of link AB.

Plan: Set the coordinate x to be the distance between A and B. Relate x to the angular position, θ . Then take time derivatives of the position equation to find the velocity and acceleration relationships.

EXAMPLE (continued)

Solution:

Relate x , the horizontal distance between A and B, to θ .

$$x = 0.2 \cos \theta \text{ (m)}$$

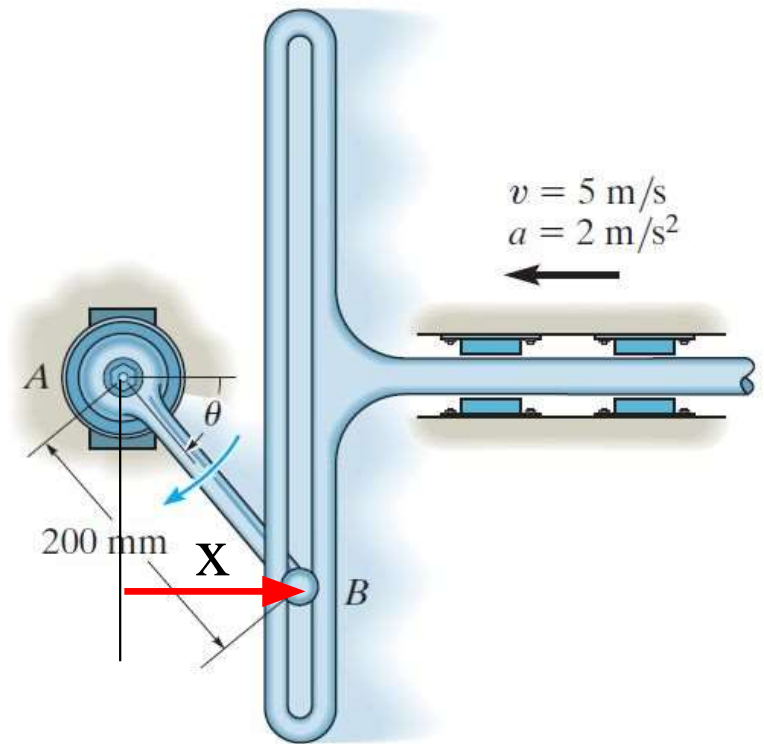
Take time derivatives of the position to find the velocity and acceleration.

$$\dot{x} = 0.2(-\sin\theta)\dot{\theta}$$

$$\ddot{x} = -0.2(\cos\theta)\dot{\theta}^2 - 0.2(\sin\theta)\ddot{\theta}$$

Notice that the velocity and acceleration of the guide rod are directed toward the negative sense of x .

When $\theta = 60^\circ$, $\dot{x} = -5 \text{ m/s}$ and $\ddot{x} = -2 \text{ m/s}^2$.



EXAMPLE (continued)

Therefore,

$$-5 = 0.2 (-\sin 60^\circ) \dot{\theta}$$

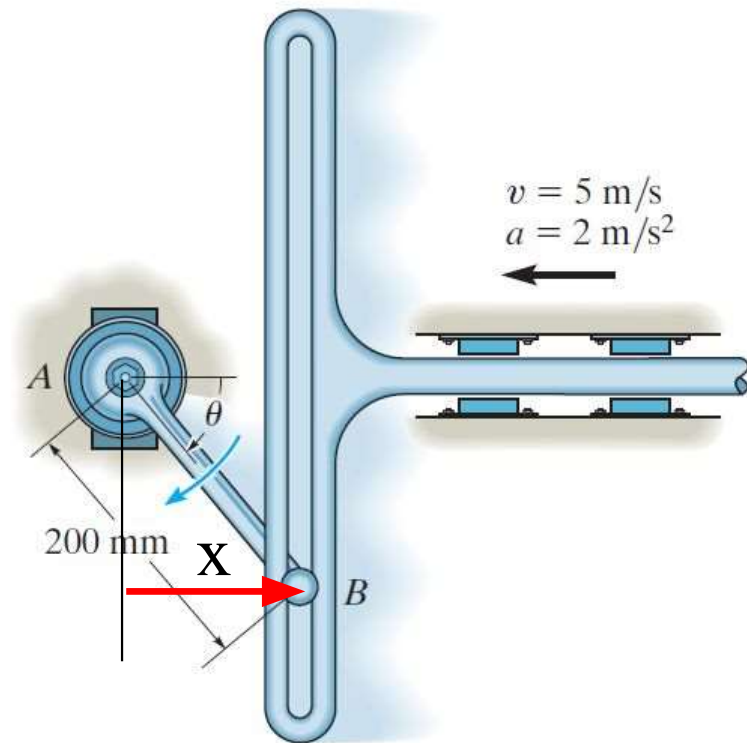
$$\dot{\theta} = \omega = \underline{28.9 \text{ rad/s}} \quad ($$

and

$$\begin{aligned} -2 &= -0.2(\cos 60^\circ) \dot{\theta}^2 \\ &\quad -0.2(\sin 60^\circ) \ddot{\theta} \end{aligned}$$

$$\begin{aligned} -2 &= -0.2(\cos 60^\circ)(28.9)^2 \\ &\quad -0.2(\sin 60^\circ) \ddot{\theta} \end{aligned}$$

$$\ddot{\theta} = \alpha = -470 \text{ rad/s}^2 = \underline{470 \text{ rad/s}^2} \quad ($$



CONCEPT QUIZ

1. The position, s , is given as a function of angular position, q , as $s = 10 \sin 2q$. The velocity, v , is

A) $20 \cos 2\theta$ B) $20 \sin 2\theta$

C) $20 \omega \cos 2\theta$ D) $20 \omega \sin 2\theta$

2. If $s = 10 \sin 2q$, the acceleration, a , is

A) $20 \alpha \sin 2\theta$

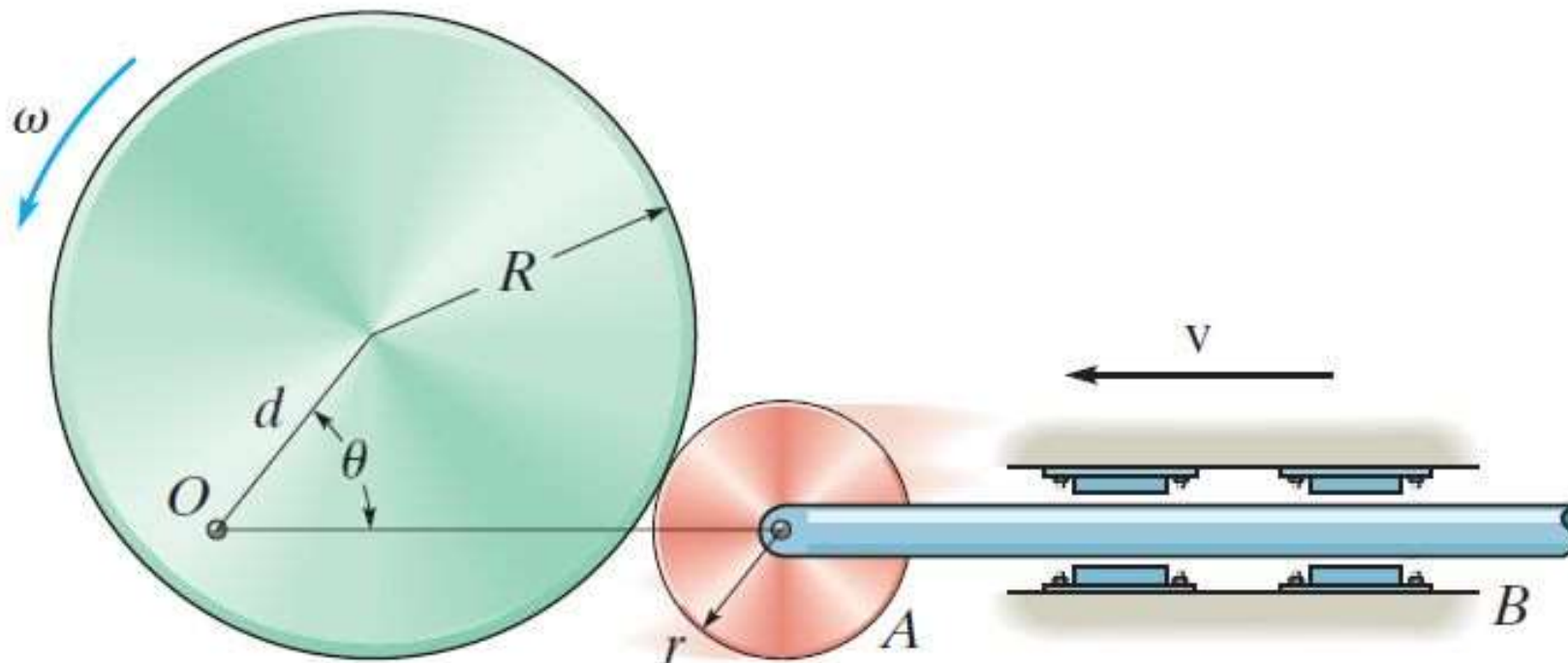
B) $20 \alpha \cos 2\theta - 40 \omega^2 \sin 2\theta$

C) $20 \alpha \cos 2\theta$

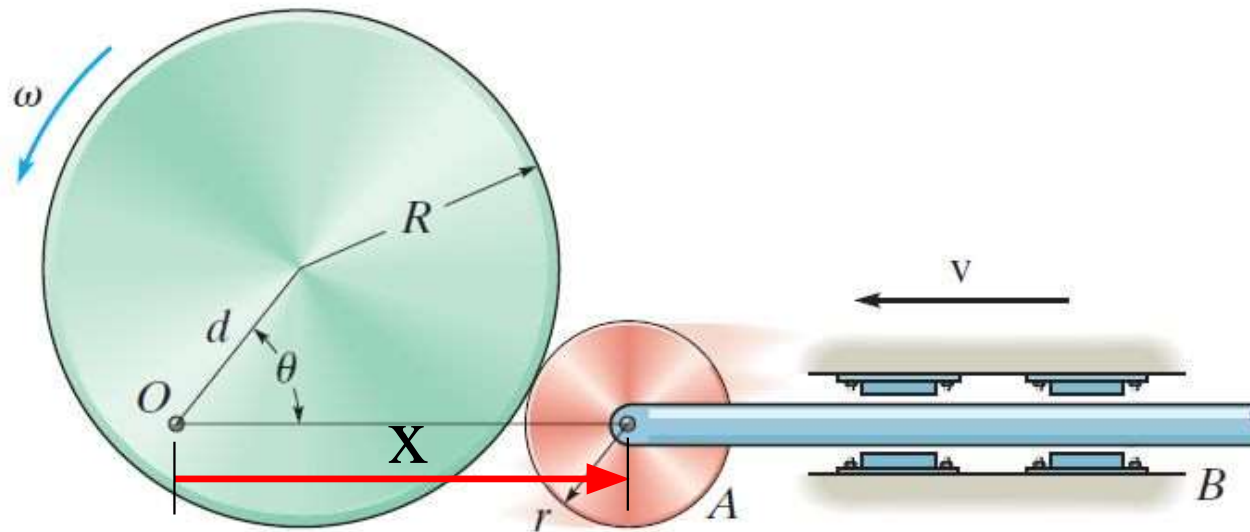
D) $-40 \alpha \sin 2\theta$

GROUP PROBLEM SOLVING

The circular cam rotates about the fixed point O with a constant angular velocity ω . Determine the velocity v of the follower rod AB as a function of θ . **(Problem No. 16-46)**



GROUP PROBLEM SOLVING



Given: The circular cam rotates about the fixed point, O with a constant angular velocity ω .

Find: The velocity v of the follower rod AB as a function of θ .

Plan: Set the coordinate x to be distance OA . Then relate x to the angular position, q . Take time derivative of this position relationship to find the velocity.

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The diagram shows a green circular disk rotating with angular velocity ω (indicated by a blue curved arrow). A point O is marked on the disk. A vector $\mathbf{X}$ is shown originating from O and pointing to a point on the disk. The magnitude of the acceleration vector is given by $a = \sqrt{(a_n)^2 + (a_t)^2}$. A red arrow points from the text "The magnitude of the acceleration vector is" to the formula. A blue arrow points from the text "To determine the motion of a point, the scalar equations" to the equations $v = \omega r$, $a_t = \alpha r$, $a_n = \omega^2 r$, and $a = \sqrt{(a_n)^2 + (a_t)^2}$ can be used. A blue arrow points from the text "The magnitude of the acceleration vector is" to the formula $a = \sqrt{(a_n)^2 + (a_t)^2}$. A blue arrow points from the text "To determine the motion of a point, the scalar equations" to the equations $v = \omega r$, $a_t = \alpha r$, $a_n = \omega^2 r$, and $a = \sqrt{(a_n)^2 + (a_t)^2}$ can be used.

$$\dot{x} = -d \sin \theta \dot{\theta} - \frac{d^2 (2 \sin \theta \cos \theta \dot{\theta})}{2 \sqrt{(R+r)^2 - (d \sin \theta)^2}} - d \sin \theta (\omega) - \frac{d^2 (\sin 2\theta) \omega}{2 \sqrt{(R+r)^2 - (d \sin \theta)^2}}$$

GROUP PROBLEM SOLVING (continued)

Note that the velocity of the rod AB is $-\dot{x}$.

$$\int_{\omega_{A0}}^{\omega_A} d\omega_A = \int_0^t \alpha_A dt$$

$$\omega_A - 20 = \int_0^t 4t^3 dt = t^4 \Rightarrow \omega_A = t^4 + 20$$

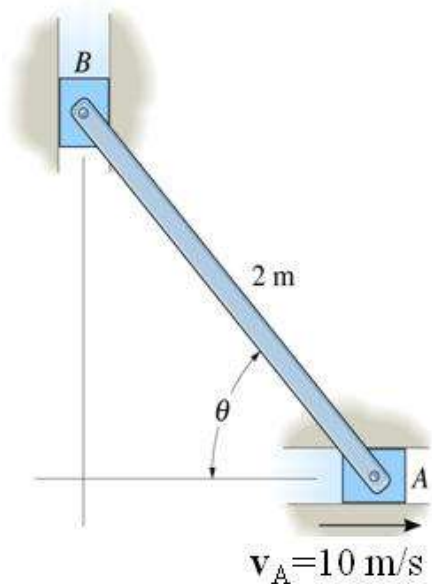
ATTENTION QUIZ

1. The sliders shown below are confined to move in the horizontal and vertical slots. If $v_A = 10 \text{ m/s}$, determine the connecting bar's angular velocity when $\theta = 30^\circ$.

A) 10 rad/s ↗ B) 10 rad/s ↗
C) 8.7 rad/s ↗ D) 8.7 rad/s ↗

2. If $v_A = 10 \text{ m/s}$ and $a_A = 10 \text{ m/s}^2$, determine the angular acceleration, α , when $\theta = 30^\circ$.

A) 0 rad/s^2 B) -50.2 rad/s^2
C) -112 rad/s^2 D) -173 rad/s^2



End of the Lecture
Let The Learning Continue